ICMP Chunk Protocol Options

Option A: 2-byte sequence + 2-byte data, capped at 1024 packets

Packet layout: ICMP identifier = 2 bytes file data, ICMP sequence = 0-1023 sequence number.

1. Read file into bytes.
2. Split file into chunks of 1024 packets × 2 bytes = 2048 bytes per chunk.
3. Send START info: filename, file size, total chunks.
4. Send chunk packets with sequence numbers 0 to 1023.
5. Receiver stores packets by sequence number.
6. Receiver ACKs only when all expected sequence numbers are received.
7. Sender waits for ACK before sending next chunk.
8. If no ACK arrives, resend the same chunk.
9. Repeat until final chunk.
10. Send END packet.

Option B: 1-byte sequence + 3-byte data

Packet layout: 8-bit sequence (0-255), 24-bit data (3 bytes).

1. Read file into bytes.
2. Split file into chunks of 256 packets × 3 bytes = 768 bytes per chunk.
3. Send START info: filename, file size, total chunks.
4. Send chunk packets with sequence numbers 0 to 255.
5. Receiver stores packets by sequence number.
6. Receiver ACKs only when all sequence numbers 0-255 are present.
7. Sender waits for ACK before sending next chunk.
8. If no ACK arrives, resend the same chunk.
9. Repeat until final chunk.
10. Send END packet.